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Enhancement on a GA-based Hotel Revenue Optimization System

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STUDENT DECLARATION

I certify that this thesis and the project to which it refers is the product of my own work and that any idea or quotation from the work of other people, published or otherwise is fully acknowledged in accordance with the standard referencing practices of the discipline.

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ABSTRACT

Revenue Management System (RMS) is a critical factor for hotels to maximize their hotel profitability. The problem here is that commercially available solutions to it remain mostly costly, opaque, and inflexible for small-to-mid sized hotels in Malaysia. This project mainly addresses the limitations to these solutions by proposing a transparent, web-based Genetic Algorithm (GA)-driven pricing optimization system that can adapt to the market based on real-world operational data. The proposed system allows the hotel managers to set an annual target revenue which is then executed by the GA to create realistic pricing strategies with factors such as seasons, room types, and market segmentation. This is done to achieve the closest ideal pricing strategies that reflects the projection of the target revenue. The system has hard business constraints such Government rate caps and has replaced the rigid quarterly models with a smart data-driven seasonality based on the historical real-world operational data from the hotels. It is implemented as a web-application via Jakarta EE 10 using Java, Apache NetBeans 27, Payara Server, and Apache Derby and the system follows the Web Development Life Cycle (WDLC) methodology. This is to achieve a smooth, user-friendly interface for all components such as configuration, execution, and result interpretation. Verification of the system has been done through functional testing which has confirmed that the system generates a feasible, pricing plans that complies within the constraints. The work demonstrates that a low-cost, interpretable, target-oriented GA optimizer can serve as a viable alternative to the commercially released RMS.

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CHAPTER 1

INTRODUCTION

This chapter provides the background and rationale for the study. It gives details of hotel revenue optimization, follows by highlighting the problem statement that led to the study. Then, this chapter continues with aim and objectives, before stating the scope and significance of the study. Lastly, it ends with a summary for this chapter.

1.1 Background of Study

A hotel's purpose is to provide a luxurious stay for the people while producing a net profit for the ones running the hotel. Hotel revenue optimization system is used to truly optimize the customer's morale and the profit garnered by the hotel. By setting room rates and allocating inventory to maximize revenue performance which are measure using either revenue per available room (RevPAR) or total room revenue (Pilar Talón-Ballester, 2022).

By using the new and improved automated revenue management systems (RMS), the process can be carried out easily by the revenue managers, distribution teams, and the analytics of the company. The systems itself cover more than just the revenue. There is strategic season planning, tactical weekly or event-based adjustments, and a rate change updates from day-to-day. Due to the nature of hotel rooms and how the business runs, an unsold room-nights could mean the deciding factor into turning a profit or making a loss (Pilar Talón-Ballester, 2022).

Therefore, even small pricings and allocation improvements will affect the financial outcomes of the company. With the hardware that is available on the market currently, automated software are created to combine the demand forecasting, price changes, and optimization algorithms such as rule based,

statistical, machine learning, or metaheuristics to recommend the pricing and allocation decisions (Pilar Talón-Ballester, 2022).

There are several options to choose from in the market when trying to find an automated RMS. These systems all show different approaches that you may have to optimize revenue. IDEaS Revenue Solutions offers an enterprise-grade RMS that integrates forecasting and dynamic pricing features into the system. The benefits of the system are game-changing but there are a few downsides such as the product's licensing costs and the complexity of the integration can be a limiting factor for smaller hotels (IDEaS, 2025). Next, the Duetto GameChanger, which is a cloud-based RMS where it supports a flexible open-pricing strategies and scenario testing. The GameChanger still has a high base level of complexity therefore it still requires a team of trained individuals to manage its configurations (Duetto, 2025) (Pilar Talón-Ballester, 2022). Other than that, Atomize is also an alternative to these two RMS. It is an AI-driven RMS which targets more mid-sized hotels with automated daily pricing recommendations. It focuses primarily on the short-term price management rather than the long term (Atomize, 2025). Lastly, PriceLabs, which is designed for short-term rentals in mind. It offers a cost-effective using a rule-based dynamic pricing for specifically small properties. The product lacks advanced hotel-specific allocation and segmentation capabilities (PriceLabs, 2025). In contrast, there are academic prototypes and metaheuristic-based approaches where the product uses Genetic Algorithms (GA). This specific style of products focuses on optimization flexibility and multi-objective modelling. The advantage of using this type of machine learning is that they can simulate complex scenarios albeit careful calibration is needed.

Genetic Algorithms (GA) is a population-based metaheuristic inspired by biological evolution. It iteratively evolves a set of candidate solutions by using mutations operators, selection, and crossovers. GA is used when the problems that are faced are nonlinear, combinatorial, or constrained by discrete decision variables. Because of these reasons, they are well adept for hotel revenue optimization systems because they can maximize revenue while minimizing

potential risks such as overbooking. They are also able to incorporate business rules within the framework (L. Haldurai, 2016).

1.2 Problem Statement

The improvement of RMS platforms over the years has proven that companies and hotels can rely on it to turn a revenue from their business. But despite the success, there are problems that comes with it. Commercialized RMS often need a high implementation cost, a high level of complexity to integrate the system, and limited transparency for smaller operators. There are many academic prototypes that remains as only a theoretical tool and lack the stability to use long-term as a product. The main issues here are insufficient modelling of local seasonality and event-driven demand changes, limited capabilities to handle market segmentations such as agents, and there is difficulty in integrating algorithmics automation with a friendly user interface for the operators (Pilar Talón-Ballester, 2022) (Chatarin Subying, 2023).

More technically, there are a few system-level challenges. Coarse seasonality modelling that relies on a fixed calendar segregation. For example, dividing a year into quarters can overlook short-term peaks and multi-scale seasonal patterns based of geographical location of the hotel. Literature supports the use of flexible, data-driven seasonality and scenario-based forecasting to improve the responsiveness of the pricing (Daniele Contessi, 2024).

Next, there are rigid pricing rules that rely on fixed multipliers fail to capture how the real-time market shifts. This emphasizes the need for human inputs into the world of automation. Other than that, the absence of market segmentation. The products that fail to account for agents or other third-party websites can misrepresent the net profitability of the business. Lastly, black-box automation without a proper human input capability can reduce the trustworthiness of the product. Managers or operators require interpretability and a level of control in the automated outputs (Oihane Gómez-Carmona, 2024).

A prototype has been developed which inherits several of these technical shortcomings that limits its full potential as a product. It divides the 365-day year into four equal quarters which means it ignores the demand variations such as public, school, or cultural holidays. Substantially, it uses a hardcoded variable into the seasonal multipliers (-0.1, 0, +0.1, +0.2) without allowing for user customizations. This in turns create a restriction in the managerial input. Furthermore, it lacks a market segmentation where it fails to account for the agents and third-party websites. These small problems can lead to inaccurate pricing and poor allocation recommendations (unpublished thesis).

To summarize, the key problem that this project is looking to solve are the current RMS solutions are often priced at a high price point, simplistic seasonal divisions, fixed percentage multipliers without user input, and lastly a market segmentation of agents and other sources modelling (Pilar Talón-Ballester, 2022) (Daniele Contessi, 2024) (Martin Petricek, 2021). This project aims to design and implement a GA-based hotel revenue optimization system that fixes these limitations by introducing a data-driven season detection, user-adjusted price change, and market-aware revenue.

1.3 Project Objectives

In order to overcome the problem in identified in the current hotel revenue optimization system, this project aims to develop a web-based hotel room pricing optimization system which utilizes Genetic Algorithm (GA). Thus, the objectives of the project are:

- a. To identify a suitable genetic algorithm approach and revenue optimization features such as appropriate seasonal changes, price multipliers that are controlled by users, and provide a market segmentation.
- b. To construct a web-based GA pricing optimization system for the hotel revenue.
- c. To evaluate the performance and functionality of the system using the hotel's dataset.

1.4 Scope of Study

The project will focus on specifically on the room aspect of the hotel's revenue. Because of this, the research in this study will exclude other sources of revenue such as food and beverage, event management, and other services. The proposed system is a dynamic system which will produce a table of optimized room prices based on factors such as seasonal demand, market segmentation, and multipliers of the room price input by the user. The scope of this research will cover three aspects of improvement as to the last system.

- First, market segmentation. The system will allow different room allocations and commission rates for agents and other third-party programs.
- Next, user-adjustable price multipliers. The earlier version of the system does not allow users to input their own ratios into the optimization room prices. The new system will allow hotel managers to manually input their own ratios into the optimization into the hotel room prices.
- Other than that, location-based seasonal changes. The system accounts for the region that the hotel is in which incorporates where and when the peak and lower seasons of the hotel will be.

The solution will deploy as a web-based system using Model-View-Controller architecture, which utilizes CSS, HTML, JavaScript for the design of the web page. PHP and SQL will be used to handle the back end of the system. The web-based system will be a more user-friendly approach to ensure that the hotel managers are able to integrate the system into their hotel seamlessly.

The dataset that is used in this project will be limited to room booking prices and the pricing data from the Royale Chulan Hotels and Resorts, which serves as a representative case study. The project does not include data from other hotel brands which may lead to inaccuracies in the output. Future extensions may include other datasets to ensure accuracy and the validity of the system are validated.

1.5 Significance of Study

Through the proposed solution, the significance towards the body of knowledge in the hotel revenue management and the advancement of Genetic Algorithm as a machine learning system by demonstrating Genetic Algorithm as a practical implementation in improving dynamic pricing accuracy and the flexibility from the management side of the system.

From an academic standpoint: This research provides an applied example of metaheuristic-based optimization in the hotel price domain, which strengthens the relationship between the research of algorithms of machine learning and the practical business operations.

From the hotelier's standpoint: This research provides hotel establishments to implement an automated optimization of room prices without relying on high prices third-party revenue management systems. The system developed with the GA provides a great amount of flexibility due to allowing user inputs and improves control over the system. As a result, hotel managers can optimize occupancy rate without sacrificing revenue.

Ultimately, this project seeks to demonstrate that GA-based optimization systems is viable as a budget-friendly, flexible, and high scalability option. This can enable them to compete in a more efficient manner in a data-driven market (Contessi et al., 2024; Gómez-Carmona et al., 2024).

1.6 Summary

This chapter describes the importance of a hotel revenue optimization system and its automation which can be beneficial in improving profitability and the efficiency of the hotel. The system being able to adjust the prices of the room dynamically, having appropriate market segmentation, and seasonal changes will directly affect the performance of a hotel's revenue. There are problems highlighted in the current systems, which include high costs, rigid seasonal modelling, fixed price multiplier, and the lack of flexibility for hotel managers

to customize their pricings. The issues can be addressed with this study's purpose of creating a web-based hotel room pricing optimization system. Genetic Algorithms will be used to dynamically calculate the optimized prices. The study is scoped into the room revenue optimization system with integrated user-adjustable multipliers, agent-based market segmentation, and geographical location-based seasonality, offers both academic value in the applied metaheuristics and the practical significance as a cost-effective hotel revenue management system for hoteliers.

CHAPTER 2

LITERATURE REVIEW

This chapter provides the literature review for the study. In this chapter, various articles from the internet that are related and relevant to the study have been reviewed. These reviews include data visualization, visualization techniques, comparison of these techniques, and similar studies. It helps to understand the general concept, terminologies and techniques in the project.

2.1 Hotel Revenue Optimization

One of the critical strategies that are integrated into the hospitality industry is the use of an efficient Revenue Management System (RMS). This can be defined as disciplined analytics in the form of an application to predict the behaviours of consumers at the micro-market level in order to optimize product availability and price of the hotel rooms to maximize revenue for the hoteliers.

2.1.1 What is Hotel Revenue Optimization?

At its core, hotel revenue optimization is the process of predicting consumer behaviour to sell hotel rooms at the most optimal price point where revenue and occupancy is maximized. According to (Zaki, 2025), it is no longer just a tool for optimizing revenue but a strategic framework for driving net profits and sustainable growth. The digitalization of the modern world is utilized at its best to create a smarter, data-driven decision-making ecosystems that integrate tailored pricing, demand prediction, and competition analysis.

2.1.2 Why is it Necessary?

In order to compete, hoteliers are demanded to maintain an operational efficiency which leads to the necessity of optimization in this age of rapid technological advancement. (Zaki, 2025) argues that a strong organization culture (OC), grounded in Social Exchange Theory (SET), is necessary to motivate employees to engage with these initiatives. Hotel that lack this

optimization will struggle to adapt to fluctuations in demands and the increasing complexity of distribution channels.

2.1.3 Who is Involved?

Traditionally, the optimization of pricing is in the domain of hotel managers. (Zaki, 2025) emphasizes that a successful optimization requires collaboration between organizational capabilities and technical know-how (TKH). This involves a “corporate revenue management team” that stays current with techniques but also relies on manpower to lessen the organization barrier to innovate.

2.1.4 When and Where is it Applied?

Continuously applying optimization through dynamic distribution networks (DN). Multi-channel distribution strategy is required to apply. For example, spanning mobile apps, OTAs, and direct booking platforms. These platforms are those that align with forecasting and dynamic pricing modules of the hotels.

2.1.5 How does it Work?

There is a step-by-step mechanism that works through enablers which has been identified by (Zaki, 2025):

- Demand Prediction (DP): Reservation trends against historical data and upcoming events are evaluated to dynamically optimize pricing.
- Tailored Pricing (TP): Strategies that are implemented for different customer segments and utilizing Best Available Rate (BAR) models.
- Competition Analysis (CA): Using and analysing competitors to assess pricing strategies and evaluating profit contribution of each market segment.

2.2 Genetic Algorithm for Optimization

The complexity of dynamic pricing can be handled by utilizing a Genetic Algorithm-based Revenue Management System (RMS).

2.2.1 What is Genetic Algorithm?

A Genetic Algorithm is a metaheuristic search heuristic that follows the process of natural selection. In recent computational studies, GAs are defined as stochastic approaches for optimization that evolve a population of candidate solutions over multiples generations (Hussain & Muhammad, 2022). It is designed specifically to solve a non-linear problem such as dynamic pricing in hotels that follow the real-time market. Traditional mathematical differentiation is unable to suffice hence the usage of GA.

2.2.2 Why use Genetic Algorithm for Hotel Pricing?

Dynamic hotel pricing is a combinatorial problem with a wide range of solutions. Recent research by Yang et al. (2025) highlights that standard heuristic algorithms will usually struggle with the nature of pricing due to its multi-objectivity. GA is preferred because of its ability global search capability where it explores the entire solution space to find “global optimum” which is the best possible solution rather than being stuck in a “local optima” which is just a good enough solution (Anshori et al., 2025).

2.2.3 Who uses GA?

Genetic Algorithm has surpassed beyond just theoretical use as it is currently being increasingly implemented in practical service sectors. It is used to optimize combat resource allocation (Zhang et al., 2025), inspection planning in manufacturing, and more in the hospitality sector for capacity planning and demand forecasting (Khalili & Mosadegh Khah, 2020).

2.2.4 How does Genetic Algorithm Work?

The process of Genetic Algorithm follows a biological evolution model with specific operators for this study.

1. Initialization: It begins with generation a random population of “chromosomes”. A single chromosome will represent a specific set of pricing multipliers for the year.
2. Fitness Evaluation: Each of of the chromosomes in the population will be tested against a fitness function. The fitter the chromosome is, the more likely it is to survive.
3. Selection (Tournament Selection): It is used to determine which pricing strategies become “parents” for the next generation and this system uses the tournament selection.
 - Mechanism: The algorithm will select a random small subset of chromosomes from the general population
 - Competition: The one with the highest fitness score will win the tournament and be selected as a parent.
 - Advantage: This selection method maintains a better diversity and prevents a single dominant solution from taking over too early which helps in the results not getting stuck in the local optima (Zhang et al., 2025).
4. Crossover: Crossover will be done on the parents once they are selected. It is a recombination between the two parents to create a more fit chromosome in the future. The system swaps parts of the pricing rules between the two parents.
5. Mutation: Mutation is applied at a rate to ensure that the algorithm will not be constrained and explores new possibilities.
6. Termination: The process will then repeat itself for a fixed number of generations and will produce the final output with the fittest chromosome. This final chromosome will be the optimal set of pricing for the hotel rooms.

2.3 Data Understanding and Analysis

To transform the theoretical Genetic Algorithm into a real practical tool for hotels to use, the development follows the Web Development Life Cycle (WDLC).

2.3.1 Raw Data from Royale Chulan

This study has utilized a full dataset from Royale Chulan Kuala Lumpur which covers the whole operational period of throughout the year of 2024. The data was provided in CSV files which details in monthly performance metrics.

The raw data is comprised of three primary components:

1. **Monthly Performance Reports (ARR 2024):** This tracks the Average Room Rate (ARR), Total Room Revenue, and Occupancy percentages through all room categories that are available in the hotel. It provides a wide range of view of how pricing decisions impacted the occupancy over the period.
2. **Market Segmentation Logs:** This part shows in detail on revenue by customer source, such as Corporate, Government, Travel Agents, and Free Independent Travellers.
3. **Seasonality Rules:** A reference dataset which defines the hotel's historical peak and off-peak periods.

A descriptive analysis of the 2024 operational dataset has revealed fluctuations in demand which are driven by Malaysia’s unique holiday calendar and seasonal events. As shown in figure 2.1, the hotel’s performance has shown its specific high demand periods that are exclusive to itself.

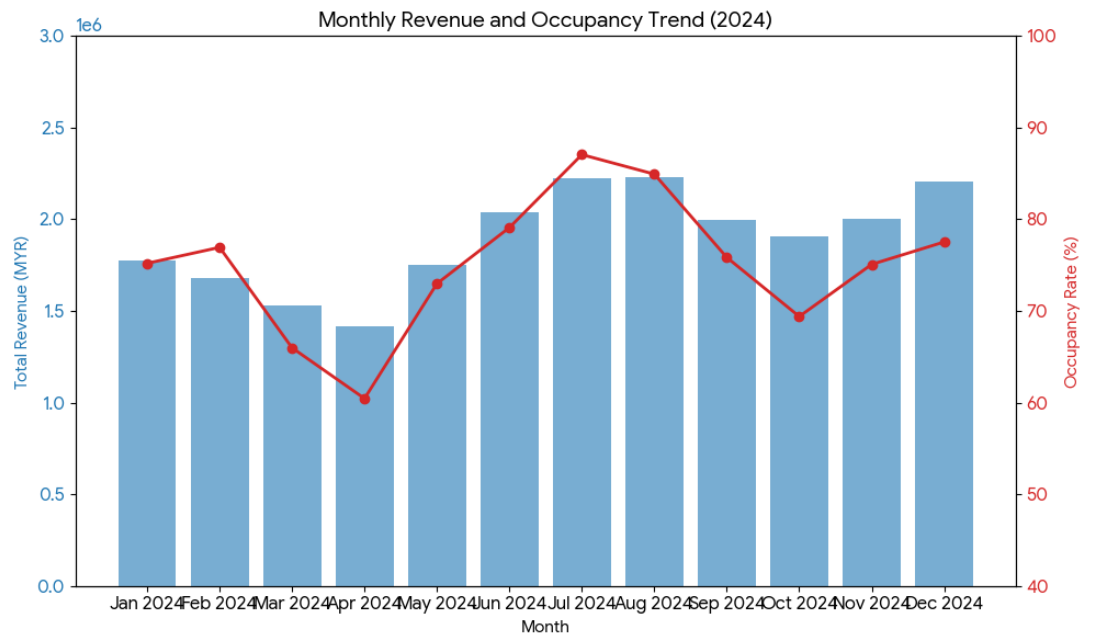


Figure 2.1: Monthly Revenue and Occupancy Trend (2024).

Key observations that can be made in the data:

- Inconsistencies through the first quartile of the year. January until February maintained a strong occupancy rates of 75.19% and 76.92% respectively while March and April struggles with 65.97% and 60.44%. This can indicate that events like New Year and Chinese New Year increased the demand during the first two months while not many events happened in March and April.
- The “Lower” season troughs identified April as the lowest point of the year with occupancy rate dropping down to 60.44% and having the lowest revenue generated with only MYR1.41 million. This low occupancy rate can be worked on to use dynamic pricing to stimulate more demands.

- Mid-Year peak in the two months of July and August representing the “Super Peak” seasons with 87.04% and 84.91% occupancy respectively. This can be explained by school holidays or cultural festivities that caused the surge in occupancy.

2.3.2 Important Features from Data

In developing the Genetic Algorithm, it requires a specific domain feature to construct the chromosomes which are the pricing solutions for the hotel. These features are extracted directly from the hotel’s raw datasets.

There are two parts on how the Genetic Algorithm based system will use the data:

1. Seasonality and Demand Patterns: The system uses the historical data in order to classify the seasonality of the months. The year will be split into four parts of seasonality depending on the occupancy rate which will be Lower, Normal, Peak, and Super Peak.
 - Super Peak (SP): July and August. These super peak months are the highlight of the hotel with 84% - 87% approximated range of occupancy. Due to the high level of demand, applying the highest price multipliers in the algorithm can lead to a significant increase in revenue.
 - Peak (P): February, June, September, November, December These months align with major school holidays and other festive seasons. Therefore, it maintains a strong 75% - 79% occupancy rate.
 - Normal (N): March, May, October, and January. These months have occupancy rates of 65% - 75% and the system could apply the base rate to balance the revenue generated while keeping adequate occupancy rates.
 - Lower (L): April. The sole month classified in the lower period with an approximate of 60% occupancy rates. This can be a trigger for the algorithm to apply a lower base rate to incentive the demographic to use the hotel’s services.

2. Room Category Contributions: Revenue contribution of the rooms can differ in type. Some of the rooms will be getting its revenue based on either High-Volume or High-Margin.

The model can differentiate between:

- High-Volume Rooms (Deluxe/Club): Optimized for maximum occupancy turnover.
- High-Margin Rooms (Suites): Optimized for rate integrity (ADR) rather than volume.

3. Market Segmentation: Using the dataset that has been provided, the analysis of identified four distinct customer groups that the system must consider.

- FIT (Free Independent Travelers): High-yield customers booking via OTAs or direct website.
- Corporate: Contracted rates that are less sensitive to dynamic pricing but provide base volume.
- Government: Regulated price caps that act as a constraint in the optimization model.
- Travel Agents: Lower net-revenue bookings due to commissions.

These features when integrated into the Genetic Algorithm system can move beyond a simple price-setting and is capable of effectively balancing between occupancy and revenue.

2.4 Similar Studies

This section serves as a review of five similar studies that are closely related to the current research in either the usage of Genetic Algorithm, the domain which is hotel revenue optimization, or both. These studies how Genetic Algorithms work, and the application of GA related to the optimization of hotel revenue. By examining scenarios that are specific to both hotels and GA, this section

can establish the methodological foundation for the proposed systems and able to highlight the research gaps.

2.4.1 Queuing-Based Hotel Capacity Planning using GA (Study 1)

This study (Khalili & Mosadegh Khah, 2020) proposes a queuing-theory-based mathematical model which integrates a two-dimensional Knapsack formulation in order to determine the optimal number and the type of hotel rooms. The queuing-theory-based mathematical model moves beyond the static demand forecasts by using the stochasticity of guest arrivals and the lengths of the stay. This is captured by simulating the hotel's reception system as an M/M/k queueing model for each room type. In this case, the guests are "customers" which arrives according to the Poisson process (exponential inter-arrival times), rooms are "servers" with service times (length of stay) which are exponentially distributed. The number of servers, k , directly corresponds to the number of rooms of the same type that are available for booking.

The two-dimensional Knapsack problem was formulated by the authors to determine the optimal value of k for each room type while also sticking to real-world business constraints. The difference between this and the classic Knapsack is that the classic only considers a single constraint while this model involves two critical limitations which are the maximum initial available space to build a hotel (S_{max}) and the maximum initial capital for construction of a hotel (B_{max}). In this case, the decision variables are the number of rooms of each type (K_j) and the constraints exist to ensure that the total required space (sum of $a_j k_j$) and the total construction cost (sum of $b_j k_j$) do not exceed the limits that has been set. This way, the authors get a formal optimization problem with hard boundaries that stems from the questions of "how many of each room type should be built?".

The objection of the two-dimensional Knapsack model is to minimize the total expected cost over a long period of time. This total cost is the sum of two opposing variables derived from the queuing model.

1. Excess Capacity Cost: The problem will happen when too many rooms are built where $k > k^*$. This is a representation of the financial loss of capital that are underutilized. The money spent on constructing and maintaining the rooms will be burnt. The cost is calculated based on the probability of having empty rooms. (πn_j where $n < K_j$)
2. Shortage Cost: This occurs when there are not enough rooms ($k < k^*$) which leads to lost revenue because guests are not able to stay in the hotel. This cost is based on the penalty of rejecting customers, calculated from the probability of all rooms being occupied (πn_j where $n > K_j$)

To solve this complex problem, a non-linear Knapsack model becomes computationally possible for realistic, large-scale problems due to the classification of it being an NP-hard problem. Methods that are exact such as branch-and-bound or dynamic programming will require exponential time which makes them impractical to use in this matter. Therefore, a modified Genetic Algorithm (GA) has been employed as a meta-heuristic solution to this problem. The GA evolves a population of potential solutions which are called chromosomes where each gene represents the number of rooms for a specific type of over multiple generations using operators such as crossover and mutation.

To ensure the efficiency of the GA, the authors have used the Taguchi method. This method is a design methodology from quality engineering to systematically tune the key parameters of the GA. Instead of a random and arbitrary selection, they conducted controlled experiments to test the levels of crossover rate, mutation rate, mutation percentage, and selection mechanism which consists of roulette wheel, tournament, and random. By analysing the results based off the average performance (signal-to-noise ratio) and the stability, they have found that the optimal combination of parameters in this case, a crossover rate of 0.9, mutation rate of 0.02 and roulette wheel selection.

The data-driven parameters tuning enhances the reliability and effectiveness of the GA. Hence the GA is justified as a “modified” GA.

With this, the model can minimize cost function which consist of excess capacity cost that is due to underutilized investment and shortage cost due to lost revenue from an unmet demand. To solve this NP-hard problem, the authors employed a modified Genetic Algorithm, with parameters using the Taguchi method.

Key strengths include:

- A realistic modelling of stochastic guest arrivals and stay durations using the M/M/k queueing systems.
- Consideration of budget and space constraints
- Use of time-value-of-money using Capital Recovery Factor for cost comparability.

However, the model here assumes distributions for arrivals and service times which may not be applicable in practice. Additionally, it does not focus on dynamic pricing, nor segmentation and only highlights the capacity planning.

2.4.2 Occupancy Forecasting Using SVR-GA Hybrid (Study 2)

This study (Anshori, Herlambang & Abu Yaziz, 2025) addresses hotel occupancy forecasting which is critical in a revenue management system. Anshori et al. (2025) integrates Support Vector Regression (SVR) with Genetic Algorithm (GA) for hyperparameter optimization. The dataset that has been used in the study involves a six-year operational data from a hotel in Surabaya, Indonesia, the SVR-GA model achieved an RMSE of 0.0186, which outperforms baseline models like ARIMA, Random Forest, and standalone SVR.

The model was developed using a longitudinal dataset which comprises of six years of monthly operational data from April 30, 2018 to May 31, 2024 from “Hotel W” which is a single establishment in Surabaya, Indonesia. The dataset has 73 observations, and it includes data such as Room Sold, Average Room Rate (ARR), Revenue Per Available Room (RevPAR), Room Revenue (RoomRev), and the target variable in this case is the Occupancy rate (%). Through a combination of Pearson correlation analysis and methods like Mutual Information (MI) and Recursive Feature Elimination (RFE), the independent features were selected to capture both linear and non-linear relationships with occupancy.

The core of the study is lies in the SVR-GA integration. Support Vector Regression is a machine learning technique that is capable or powerful modelling complex, non-linear relationships regarding data. Even so, the performance of SVR is highly sensitive to the values of hyperparameters which are the Cost (C), Epsilon (ϵ), and Gamma (γ). The authors employed a Genetic Algorithm to find the optimal set of these said parameters. The problem is now transformed into an optimization task where the GA is now searching for a combination of parameters that minimizes the prediction error. The error specifically the Root Mean Square Error – RMSE.

The GA process is as follows:

1. Initialization: A population of 10 candidate chromosomes are created. Each is a triplet that represents the potential values for the three SVR hyperparameters which are the Cost, Epsilon and Gamma. The values are randomly generated but with constraints which are the predefined ranges.
2. Evaluation: Each of the chromosome’s fitness is evaluated by training an SVR model with the specific hyperparameters on the training data which then will calculate the RMSE on the test data. A lower RMSE means to a higher fitness score.
3. Selection: A tournament selection mechanism is used in the study. Three chromosomes are randomly chosen from the population and

the one with the best fitness score is selected as a parent for the next generation of the chromosomes.

4. Crossover: Two selected parent chromosomes will then exchange parts of the genetic information added with a probability of 0.5 to produce two new child chromosomes. The successful traits of the parents will then be combined.
5. Mutation: To maintain diversity and to prevent the premature convergence to a local optimum, mutation operator is applied with the probability of 0.2. This operation will randomly alter one or more of the hyperparameters value in the child chromosome by a tiny amount.
6. New Generation & Iteration: The offspring from the crossover and the mutation will then form the next generation. Step 2-5 will be repeated for a fixed number of generations. The algorithm will then return the hyperparameters with the best fitness scores among the chromosomes.

Figure 2.4: Research Methodology Flowchart for the SVR-GA Hybrid Model.

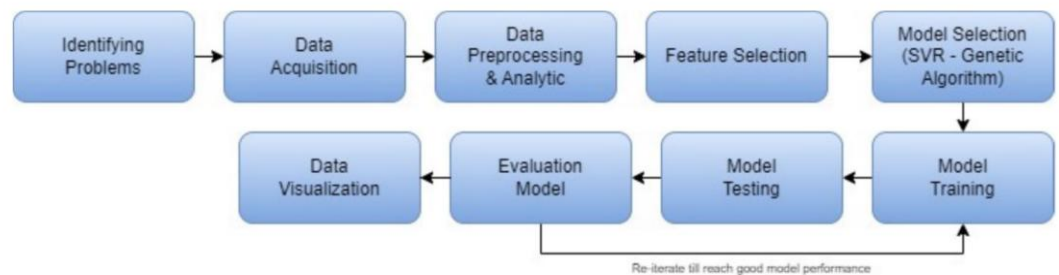


Figure 2.4

The SVR model that has been trained with GA-optimized parameters was tested using k-fold cross-validation to ensure its robustness.

The results of the SVR-GA model achieved a remarkably low Root Mean Square Error (RMSE) of 0.0186, which represents a significant leap over other

baseline models. It outperformed SVR (RMSE = 0.1254), Random Forest (RMSE = 0.0832), ARIMA (RMSE = 0.0957), and LSTM (RMSE = 0.0479). This shows the level of accuracy that the hybrid approach of SVR-GA can achieve.

The contribution in the study involves:

- A demonstration of GA's effectiveness in tuning SVR parameters.
- Validation in the application of the model.
- An emphasis on the accuracy of prediction for strategic pricing.

The limitations in this study include the lack of variation in the dataset as it relies solely on a single hotel's data. This limits the generalizability of the results.

2.4.3 Dynamic Pricing and Revenue Optimization in Hospitality (Study 3)

Bale & Afiniki (2024) investigate the impact of dynamic pricing strategies which involves time-based, demand-based and yield management on the revenue optimization in the sector of hospitality. The study uses a variety of methods such as surveys, interviews, and regression analysis. They have found that dynamic pricing significantly boosts revenue, especially during the peak seasons and in regions that are economically stable.

The quantitative aspect of this study was centered around a structured survey questionnaire which has been given to 250 participants across hospitality establishments in Plateau State, Nigeria. This includes hotels, restaurants, resorts, and event venues. After data cleaning has been done on the survey respondents, the dataset finally consists of 219 valid responses. The survey used a 5-point Likert scale to measure attitudes and perceptions on things such as the usage of dynamic pricing in hospitality, its effectiveness on revenue performance, and the factors that are directly influencing the decisions of dynamic pricing. Questions were designed to question aspects such as:

- The frequency and type of dynamic pricing
- Perceived changes in revenue and occupancy rates when dynamic pricing is implemented
- The influence of competition in the market and the operational constraints on the flexibility of the pricing.
- The part that technological infrastructure plays in enabling dynamic pricing.

This primary survey data was added alongside secondary data that is collected from industry reports, financial records, and online databases to provide insights into the current state of the market and economic conditions.

To statistically validate the relationship between dynamic pricing and business outcomes, the authors has conducted a regression analysis. The analysis is a multivariate analysis which examines the relationship between the variables that are dependent, which is Revenue Optimization and several independent variables, which are Pricing Strategies (PS), Revenue Performance (RP), Use of Dynamic Pricing Techniques (DP), and Factors Influencing Pricing Decisions (FPS). The analysis revealed a statistically significant positive relationship between the use of dynamic pricing techniques and revenue performance. This shows that there is strong quantitative evidence to back up their effectiveness.

Other than that, there are in-depth interviews with twenty key stakeholders which includes hotel managers, revenue managers, and restaurant proprietors. The interviews were conducted over a three-week period. Additionally, there were group discussions that are held with other hospitality professionals. The sessions are to explore deeply into practical challenges, managerial decisions, and experienced knowledge that are related to implementing dynamic pricing. The questions in the interviews involve:

- How to determine when to adjust prices.

- What are the factors that influence the pricing decisions.
- How do customers reach to the price changes and how to communicate it transparently to the customers.
- What are the biggest obstacles in implementing the dynamic pricing.

By using these three methods, triangulation of the survey with the interview insights along with the statistical analysis, it can provide a more comprehensive view of the impact of dynamic pricing. The study has found that dynamic pricing boosts revenue, specifically during the peak seasons and in regions where the economy is stable.

Key insights:

- The behaviour of consumers will influence the success of the pricing. There are factors such as mobile app usage, and loyalty program participants that influence the customer's perception of dynamic pricing. Tech-savvy customers who book through mobile applications are more used to fluctuating prices.
- Communication and transparency in the change of price will enhance the acceptance of consumers.
- A good technological infrastructure and the accuracy of data are needed for effective implementation

While this study excels in the managerial aspect, the study is non-algorithmic as it does not involve any Genetic Algorithm or any machine learning. It evaluates adoption and perception of consumers rather than proposing or testing an optimization algorithm.

2.4.4 Optimizing Booth Pricing with a Staged Crossover Hybrid Genetic Algorithm (SCHGA) (Study 4)

This study by Yang et al. (2025) addresses a critical challenge in the current exhibition industry. The problem here is that how to optimize booth pricing and its layout to maximize revenue for the organizers while at the same time accommodate to the diverse demands of exhibitors. The traditional fixed pricing model fails to account for factors that are important such as the booth location, competition between the exhibitors, and varying price sensitivities across brands. The authors have proposed a mathematical model for booth pricing optimization that uses economic theories which includes hedonic pricing, monopolistic competition, and price discrimination. The methods used are combined with real-world data. The core innovation is in the development of Staged Crossover Hybrid Genetic Algorithm (SCHGA) which is designed to solve a complex optimization problem in a more efficient manner.

The proposed mathematical model is comprised of four main components:

1. Base Revenue: Time-dependent base price (pricebase) which is adjusted by a coefficient (P_t) which will decrease over the registration period to reflect early-bird discounts.
2. Location Value: Premium pricing in high-traffic areas. It is quantified by using the variable “location score”.
3. Competition Penalty: Penalty term which will reduce the effective price when competitors are placed close together. This will mitigate internal market friction and preserve value for all the participants.
4. Price Sensitivity Adjustment: Adjustment based on an exhibitor’s sensitivity to price. Smaller brands are charged lesser while larger brands can be charged a premium price due to their lower sensitivity to price.

The model’s crucial aspect is the method for quantifying inter-exhibitor competition. The authors introduced a concept called “competitive semantic similarity” which calculates a competition score that are based on three weighted dimensions of company information. The dimensions are company

name, industry category, and business scope. The weights for each respective dimension are 0.1, 0.5, and 0.4. The weights are determined through an expert evaluation that emphasizes a greater importance towards the importance of industry and business scope in terms of identifying true competitors. The semantic similarity score is then divided by the Manhattan distance between the two booths which will create the final competitiveness score that increases based on both the product similarity and spatial proximity. This approach ensures that that competitors are not placed right next to each other.

Assigning N exhibitors to M booths while following constraints on space and maximizing the complex objective function to solve the multi-objective optimization problem is computationally challenging. The standard heuristic algorithms such as Particle Swarm Optimization and Ant Colony Optimization were tested. It was found that they are less effective while a basic Genetic Algorithm showed promising results. The authors then identified a key limitation in a traditional Genetic Algorithm specifically in the crossover mechanism. The way that the conventional GA treats the booth's coordinate (x,y) as an indivisible unit during the crossover means that both values are swapped together from one parent to their children. This can cause disruption in a solution that has one coordinate that is already optimized while the other still needing more optimization. This problem can lead to a premature convergence and a reduced search efficiency.

To address this issue, the authors developed a Staged Crossover Hybrid Genetic Algorithm (SCHGA). The key innovation in this algorithm is the coordinate-point crossover mechanism. Instead of swapping the entire coordinate (x,y) pairs, the algorithm lets the crossover point to occur within the coordinate pairs. This means that the x -coordinate of a new offspring can come from one parent, while the other coordinate originates from another parent. This can enhance convergence speed and the quality of the solutions produced.

The SCHGA are operated in two different stages:

1. Early Stage (Generations 1-100): The algorithm uses a simulated annealing mechanism with the standard GA crossover that swaps the coordinate pairs as whole. This will aid in the population's exploration of solution space.
2. Late Stage (Generations 101+): After the 100 generations mark, the algorithm will then switch into the coordinate-point crossover mechanism. The solution at this stage will be close to an optimal region and fine-tuning can allow for precise adjustments which leads to an accelerated final convergence.

The SCHGA was tested using real data from the Shenzhen Convention and Exhibition Centre. Experiments are compared against advanced algorithms such as PAGA and MAPSO and other baseline algorithms like GA, PSO, ACO, and DE. The SCHGA achieved the highest maximum, average, and minimum fitness values and the lowest standard deviation, which shows its superiority compared to other algorithms. The solutions that are produced from the algorithm are in exceptional quality.

While the study presents an effective solution, it acknowledges limitations, which includes its reliance on data from only a single exhibition centre and all assumptions that based off this dataset alone. Future improvements could extend the model into more data sources.

2.4.5 Dynamic Pricing Optimization Using Sim-Learnheuristic Framework (Study 5)

This study (C-Sánchez & Gomez, 2025) addresses the challenge of applying dynamic pricing in hotel revenue management and proposes a novel Sim-Learnheuristic model that hybridize simulation, heuristic search, and machine learning to optimize daily room pricing decisions. Due to limitations of traditional deterministic and stochastic approaches such as their rigidity and lack of adaptation to real-time market dynamics, the author introduced a methodology that uses a blend that continuously learns from historical booking patterns while efficiently explore pricing strategies through a biased randomization. The framework is benchmarked against deterministic and sim-heuristic models that uses real-world hotel data and demonstrates a better computational efficiency and competitive revenue performance.

The development of the model utilized an open-access dataset that was derived from a city-type hotel in Lisbon, Portugal (Antonio et al., 2019) which compromised of approximately 79,000 individual bookings that was recorded between 2015 and 2017. To ensure temporal completeness, the study has isolated the full calendar year of 2016 as the primary analysis period. The raw dataset has undergone aggregation to transform the data into a more structured format that is suitable to be utilized in the optimization modelling.

The key variables that are extracted include:

- Average Daily Rate (ADR): Calculated as total lodging revenue divided by total nights stayed, serving as the reference price point.
- Temporal markers: Year, month, and day of arrival for each reservation
- Lead time: Days elapsed between booking placement and check-in date
- Room type: Categorical classifications of accommodation tiers
- Length of stay: Duration of guest accommodation that is derived from check-in and check-out dates

The target variables are formulated as the aggregated demand count for specific time pairs which is (a,b) where a represents the check-in date and b represents the check-out date. This utilization of paired aggregation strategy captures the interdependency of multi-night stays.

The study developed and benchmarked three distinct models where each of them represented a different approach to dynamic pricing optimization of hotel rooms.

Deterministic Model (DE):

This deterministic formulation treats the pricing problem as a straightforward mathematical programming exercise with complete information. Mixed-Integer Programming (MIP) was used to maximize revenue subject to capacity constraints, where demand is modelled as deterministic function of price. The objective of this model is to maximize total revenue from both regular bookings and stay-over extensions. Capacity constraints in this model ensure total demand never exceeds available rooms. This way, global optimum which is the absolute maximum revenue achievable under perfect circumstances are calculated or predicted. However, the complexity of the computation grows exponentially with problem size. It requires over 8 hours to solve a 365-day horizon for a 110-room hotel. It is also unable to adapt to real-time market changes once the solution is computed.

Sim-Heuristic (SH):

This sim-heuristic model maintains the advantage of perfect future information but replaces exact mathematical optimization with a biased randomization heuristic. The method took samples from a discrete set of prices that are management-approved. The selection probabilities are skewed towards options that are historically successful while still preserving exploration capacity. The algorithm evaluates each of the candidate price using simulation

and calculate the fitness based on revenue potential and iteratively refines the strategy of selection. Local optimum solution is calculated within 2-4 minutes, and the model achieved a 65-85% of the deterministic optimum. However, the limitations are still like the deterministic model where it is unable to adapt to real-time market or learn from new data.

Sim-Learnheuristic (SLH):

This proposed sim-learnheuristic model represents a change in the model's rigidity by incorporating predictive learning into the optimization loop. It consists of three components that have been integrated in continuous feedback:

- **Black Box Market Simulator:** This component uses a non-parametric demand generator which replicates real market behaviours using the historical data of Lisbon. It advances bookings, walk-ins, and stay-overs while also incorporating price elasticity and seasonality effects.
- **White Box Demand Forecaster:** This is a two-stage Random Forest ensemble that addresses data sparsity. Stage 1 is a classification to predict binary occupancy for each date pair. Stage 2 applies regression to forecast demand magnitude when Stage 1 predicts a positive in occupancy.
- **Heuristic Price Selector:** This component uses a biased randomization to explore a discrete price space that has been guided by forecasting rather than the actual demand. The probability distribution updates dynamically based on the predictions that have been made by the white box and creates a learning-while-searching mechanism.

The experiment design systematically evaluated all three models across varying time horizons which are 10, 30, 60, 180, and 365 days. Hotel capacities of 50, 80, and 110 rooms with the elasticity fixed at -1.5 and α at 0.7. The sim-learnheuristic approach has shown significant time savings when compared to the other models. For the 365-day horizon with 110 rooms, the deterministic model required over 9 hours while the sim-learnheuristic completed in only

approximately 3 minutes. The sim-heuristic also performed quite similarly at 106 but lacks real-time update. In terms of the revenue performance, the deterministic model achieved the global optimum of €1,746,125 for the 365-day/110-room scenario. The sim-heuristic reached €1,344,703 (77% of optimal), while the sim-learnheuristic achieved €846,950 (49% of optimal) from a cold start. While it seems that the sim-learnheuristic has a low percentage, the relative error decreased significantly with shorter horizons. For example, 10 days with 50 rooms, SLH achieved only 14.75% error versus the sim-heuristic benchmark, demonstrating strong forecast accuracy even with limited training data. Moreover, performance has shown to improve consistently given larger capacities such as 110-room hotels showing 23-37% error versus 42-58% for 50-room properties. The sim-learnheuristic has a trade-off in the theoretical optimum for dramatic change in time taken for calculation and its ability to adapt to real-time market.

In conclusion, the research establishes sim-learnheuristic as a practical, computationally efficient framework for dynamic pricing in hotels to overcome the problem of rigidity of traditional methods.

2.4.6 Comparison of Similar Studies

Table 2.2 shows the comparative table for each of the similar studies that are researched. This comparison table are made for summarizing the contents of the studies.

Table 2.2: Comparison of similar studies.

Studies	Study 1	Study 2	Study 3	Study 4	Study 5
Author (Year)	Khalili & Mosadegh Khah (2020)	Anshori et al. (2025)	Bale & Afiniki (2024)	Yang et al. (2025)	C-Sánchez & Gomez (2025)
Methodology	GA + Queueing Theory (M/M/k)	Hybrid SVR-GA	Survey & Regression	Staged Crossover Hybrid GA (SCHGA)	Sim-Learnheuristic
Focus Area	Capacity Planning	Forecasting	Managerial / Adoption	Booth Pricing	Computational Efficiency
Key Contribution	Optimized the number of rooms to build using a 2D-Knapsack formulation.	Used GA to tune hyperparameters for Support Vector Regression, achieving high forecasting accuracy (RMSE 0.0186).	Validated that dynamic pricing statistically improves revenue and identified key managerial challenges.	Introduced "Coordinate-Point Crossover" to solve spatial pricing problems for exhibitions.	Proposed a fast "learning-while-searching" model that outperforms exact methods in speed (3 mins vs 9 hours).
Limitation / Research Gap	Focused on long-term construction (strategic) rather than daily dynamic pricing (tactical). Assumed fixed arrival distributions.	Solved the forecasting problem but did not extend to the optimization of prices based on that forecast.	Purely descriptive/quantitative social research; offered no technical solution, algorithm, or software tool.	Context-specific to exhibitions (spatial/location-based) which differs from the temporal/seasonal nature of hotel room pricing.	The model is a "Black Box" simulation that lacks transparency, making it difficult for managers to intervene or adjust multipliers manually.

From Table 2.2, Genetic Algorithm (GA) and hybrid variations of the algorithm is proven as suitable for optimizing hotel revenue.

This is justified by the comparison criteria of computational efficiency, adaptability to non-linear constraints and NP-hard problems. (Bale & Afinki, 2024), provide managerial insights from the hotelier's point of view and exact mathematical models like (Khalili & Mosadegh Khah, 2020), offer precision but lacks the flexibility to handle real-time market demands in a small timeframe.

In contrast, Genetic Algorithms has shown that it has its own capabilities to solve NP-hard combinatorial problems such as assigning the right price to the right room type to the right customer segment. The balance in GA systems make it the most suitable choice for the project.

2.5 Summary

This chapter has covered the framework in theory and the context that is required to develop a Hotel Revenue Optimization System. It began by the definition of the core principles of revenue management which emphasizes on the need of dynamic pricing and demand prediction strategies to maintain operational efficiency and profit maximization in the hospitality industry. The chapter explains in detail about the machine learning model Genetic Algorithm (GA) which has been chosen as the metaheuristic approach. This explains about its biological inspiration and specific operators involved in the model like the selection method, crossover, and mutation. This model has the capability of solving complex, non-linear problem where traditional methods fail.

Furthermore, real-world operational data was analysed from Royale Chulan Kuala Lumpur (2024) and the critical domain features such as the seasonality, room category yield, and market segmentation constraints. Lastly, a review of five similar studies was conducted, ranging from queuing theory for capacity planning to hybrid machine learning models for forecasting. The analysis has shown the research gaps in methodologies that are existing and is used to justify the usage of Genetic Algorithm (GA) to develop the revenue system. GA is shown to be the most suitable technique to balance computational efficiency and the adaptability that is required for the hotel.

CHAPTER 3

METHODOLOGY

This chapter conveys the flow of the project method from the beginning phase until the complete project report. The entire flow of this study such as method used and how it is implemented is explained here. Research methodology is a systematic way for solving the research issue by legitimately embracing different advances. Methodology comprehends not just the final system but also the process itself.

3.1 Software Development Methodology

This study has taken up Web Development Life Cycle (WDLC) as its foundational software development methodology. WDLC is a method that utilizes a structured, phase-oriented framework that is specifically designed for creation, deployment, and evaluation of web-based applications such as our own. The difference between WDLC and other general-purpose models such as Waterfall or Agile methodologies that prioritizes rigid sequential process or rapid iteration respectively is that WDLC offers a more balanced approach that integrates a user-centered design, modular system architecture, and lastly iterative validation. These are all essential into the process of developing a full-fledged system.

The WDLC model consists of four core phases:

1. Requirement Analysis.
2. System Design.
3. Implementation.
4. Verification and Testing.

Each of the phases in WDLC uses the outputs of the previous steps while simultaneously allow for controlled feedback loops to refine specifications. The phased progression is an added advantage because it ensures that the technical robustness and operational usability are achieved at the same time.

This addresses a critical gap which has been identified in the last form of the GA-based hotel revenue management system that functions as “black boxes” with limited managerial controls.

Using WDLC can be justified by the nature of the proposed system which is a web-deployed, data-driven decision support tool that is intended for hotel revenue managers. WDLC is a method that emphasizes specific points that aligns with the project such as prototyping, stakeholder interaction, and front-end and back-end co-design. Moreover, WDLC accommodates the use of advanced computational techniques. For example, metaheuristic optimization within a practical software. Genetic Algorithms are originally from evolutionary computation theory (Hussain & Muhammad, 2022) and to implement a successful deployment in the hospitality context, it requires more than just algorithmic correctness. It also requires a seamless embedding into a feasible workflow. WDLC can provide as the scaffolding for this specific integration where the GA operates as the “Model” in the MVC-architecture.

In summary, WDLC works as a checklist and the bridge to theoretical optimization (GA), domain knowledge (hotel revenue optimization), and user experience (web interface).

3.2 Project Framework

This section describes the project framework for this project (see Table 3.1). The phases in the project framework will follow the phases in the web development life cycle (WDLC). The objectives are mapped to the phases in the project framework.

Table 3.1 Project Framework

Objective	Phase	Activity	Outcome
a. To identify a suitable genetic algorithm approach and revenue optimization features such as appropriate seasonal changes, price multipliers that are controlled by users, and provide a market segmentation.	Requirement Analysis	1. Review on Hotel Revenue Optimization concepts and commercial RMS limitations. 2. Review on similar systems that utilizes Genetic Algorithm operators from literature. 3. Review on modelling the seasonality techniques based on the Royale Chulan 2024 dataset. 4. Review on the market segmentation rules and the net-revenue constraints.	1. Problem Statements 2. Objectives 3. Scope 4. Significance 5. Comparison tables
b. To construct a web-based GA pricing optimization system for the hotel revenue.	System Design	Design on the following diagram: 1. System Architecture 2. Process Flow 3. Use Case Diagram 4. Use Case Description 5. Algorithm Design 6. User Interface Design	1. System Architecture Diagram 2. Process Flow Diagram 3. Use Case Diagram 4. Five tables of Use Case Description 5. NB: Algorithm Pseudocode 6. Three User Interface Design Mockups
	Implementation	1. Data Setup 2. Software Requirements 3. Front-end 4. Back-end 5. Database Integration	1. Functional Web Application 2. Complete Source Code 3. Preloaded Derby and Database Schema and Sample Data.

			4. Configuration Files 5. README.md with instructions.
c. To evaluate the performance and functionality of the system using the hotel's dataset.	Verification and Testing	1. Test Case Generation 2. Selection of Testers 3. User Acceptance Test	1. Functionality Test Results 2. UAT Results 3. Performance Metrics Report 4. Bug Log and Resolution Summary.

The following table shows the structural mapping of the project's core objectives in relation to the phases of the Web Development Life Cycle (WDLC). It has been shown in detail the specific activities that are performed alongside the tangible outcomes that are generated from each stage of the cycle. This framework ensures that all development tasks will have its own contribution into fulfilling the three main objectives of the study.

3.3 Phase 1: Requirements Analysis

This phase is to gather all the information regarding the development of a web-based Genetic Algorithm (GA)- driven hotel revenue optimization system in the application development. To further understand this, research on hotel revenue management system, metaheuristic optimization techniques, and user-centered web design has been done and further focused into three core areas which are the limitations of commercial revenue management systems, the application of Genetic Algorithm in dynamic pricing, and lastly, the need of managerial control over the pricing decisions. Method of gathering this information is through online publications such as journal, articles, proceedings.

The preliminary study started with a comprehensive literature review that consists of five recent studies ranging from the year 2022 until 2025. It is comprised of AI-based hotel pricing systems, which highlighted gaps in the market such as high licensing cost, black-box automation, rigid seasonal modelling, and a lack of market segmentation. In effort to visualize the findings, an in-depth analysis of Royale Chulan Kuala Lumpur's 2024 operational dataset was conducted. Monthly occupancy, Average Room Rate (ARR), and market segment contributions (FIT, Corporate, Government, Travel Agents).

Simultaneously, technical feasibility was assessed and suitable hardware and software requirement for development and demonstration was found.

Table 3.2: Software Requirements

Category	Specifications
Development IDE	Apache NetBeans 27
Programming Language	Java (JDK 24)
Web Framework	Jakarta EE 10
Application Server	Payara Server 6.2025.10
Database	Apache Derby (embedded mode)
Frontend Technologies	Jakarta Server Pages (JSP), JSTL, Bootstrap 5
Deployment Environment	Local machine (Windows 11, Intel i7-11800H, 16GB RAM)

After gathering all the necessary information, further reading was done with the objective of achieving a better understanding regarding the study. The problem statement can be identified as well as the objectives of the project. Specifically, the findings confirmed that existing RMS solutions are limited by the high cost of licenses and the lack of transparency for small-to-mid sized

hotels in Malaysia. On the other hand, academic prototypes lacked the ability to handle real-world data integration and managerial control mechanism.

The outcomes of this phase have produced the problem statements, objectives, scope and the significance of the project.

3.4 Phase 2: System Design

This phase aims to design the proposed system which includes the system architecture, process flow, use case diagram, use case description, user interface design.

3.4.1 System Architecture

The proposed system architecture will adopt the Model-View-Controller (MVC) architectural pattern. This is a standard in Jakarta EE web development which focuses on the separation of concerns, enhancing maintainability, and supports iterative refinement. The limitations that are stated in “black-box automation” in commercial RMS are overcome with the selection of using MVC as it decouples the computational logic from the user interaction. Figure 3.2 shows the general workflow of a Model-View-Controller architecture.

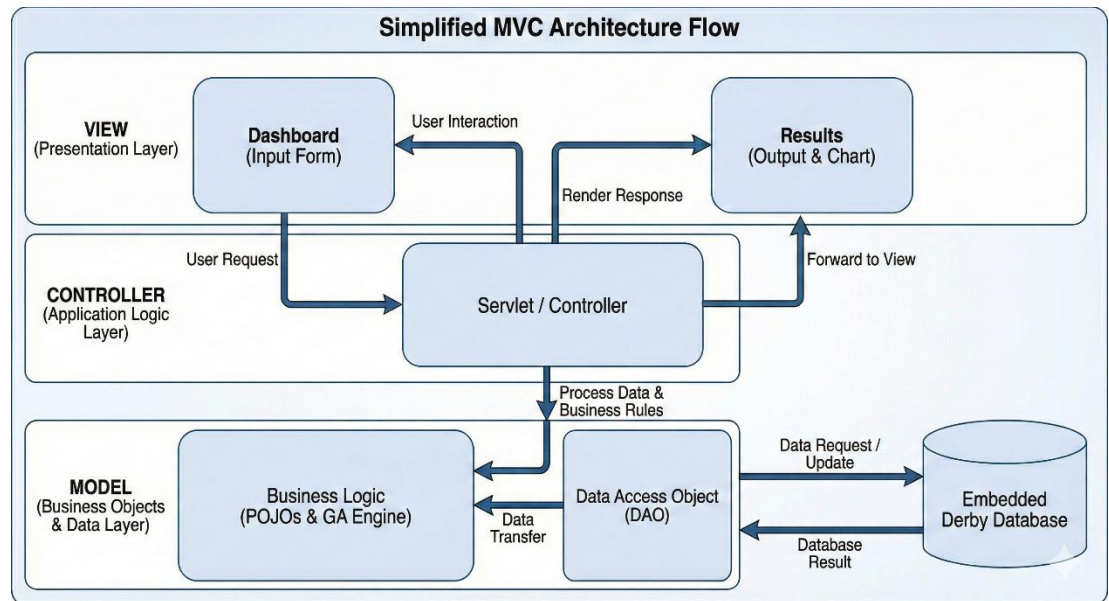


Figure 3.2 System Architecture

These are the rough contents of what the three layers will contain:

- **Model Layer (Java Classes + Derby JDBC)**
 - `SeasonalityClassifier.java`: Classifies months into 4 different classifications which are Lower (L), Normal (N), Peak (P), Super Peak (SP) that is based on occupancy thresholds from the dataset
 - `MarketSegment.java`: Enum or class defining four segments with rules around the net-revenue. (e.g., GOVERNMENT enforces the max multiplier = 0.8)
 - `RoomCategory.java`: Differentiates between high-volume vs high-margin rooms.
 - `GeneticAlgorithm.java`: A pure java implementation of the Genetic Algorithm engine used to determine room prices.
 - `DerbyDAO.java`: Handles JDBC connections to the database.
- **View Layer**
 - `Dashboard.jsp`: Main interface for the users.

- Results.jsp: Display a table of the optimized pricing for rooms based of target revenue.
- Controller Layer
 - RunGAServlet.java: Core controller which handles multiple functions.
 - Reads user inputs
 - Fetches data from database tables.
 - Executes the GeneticAlgorithm.java.
 - Stores results.
 - Forwards into results.jsp.
 - LoginServlet.java, ProfileServlet.java: Handles user authentication and user profiles.

Due to the structure of the architecture, it is fully traceable. Every business rule is enforced in Java classes, and it is not hidden behind in SQL logic nor UI logic. This addresses the “black box” critique of commercial RMS.

3.4.2 Use Case Diagram

Figure 3.3 shows the common use cases for the proposed hotel revenue optimization system.

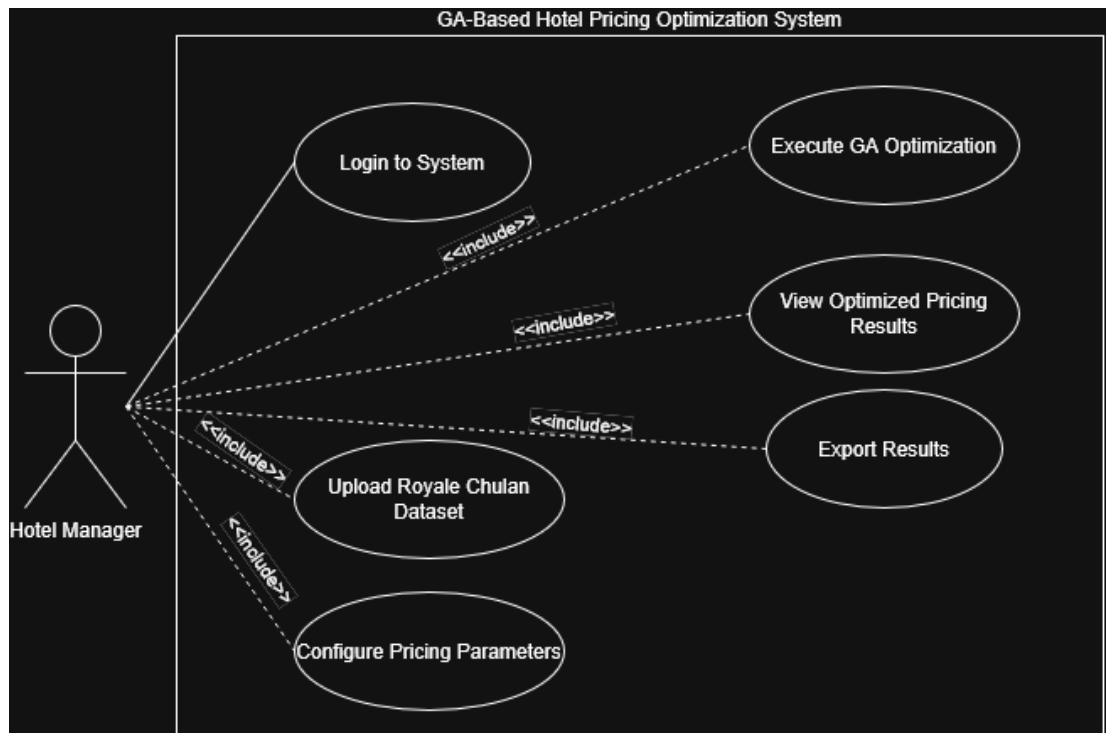


Figure 3.3 Use Case Diagram

From Figure 3.3, there are six use cases in movie recommender system. As an administrator, the system allows to login to the system, execute the GA optimization, view optimized pricing results, export results, upload the hotel's dataset, and configure pricing parameters.

Table 3.1 shows the use case description for add movies as an administrator.

Table 3.1 Use Case Description for Login to System

Use Case ID	UC 001
Use Case	Login to System
Purpose	To authenticate the Hotel Manager before accessing system functions.
Actor	Hotel Manager
Trigger	User selects “Login” button on homepage.
Precondition	None (system is accessible to unauthenticated users).
Main Flow	1. User enters username and password. 2. System validates credentials against internal user table. 3. On success, user is redirected to dashboard. 4. Session is established for subsequent use cases.
Exception Flow	1a. Invalid credentials → Display error message “Invalid username or password.” 2a. Empty fields → Display “Please fill all fields.”

Table 3.2 Use Case Description for Upload Royale Chulan Dataset

Use Case ID	UC 002
Use Case	Upload Royale Chulan Dataset
Purpose	To import historical hotel data (CSV) for GA optimization.
Actor	Hotel Manager
Trigger	User clicks “Upload Dataset” after logging in.
Precondition	User must be logged in.
Main Flow	1. User selects CSV file from local machine. 2. System validates file format (headers: month, occupancy, ARR, segment, etc.). 3. Data is parsed and stored in Apache Derby database tables. 4. System displays summary: “Dataset uploaded successfully. 12 months loaded.”
Exception Flow	1a. File not selected → Display “Please select a CSV file.”

	2a. Invalid format → Display “File structure mismatch. Please use Royale Chulan 2024 template.” 3a. Database error → Display “Failed to save data. Please try again.”
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Table 3.3 Use Case Description for Configure Pricing Parameters

Use Case ID	UC 003
Use Case	Configure Pricing Parameters
Purpose	To allow the manager to set global/segment-specific price multipliers and lock segments.
Actor	Hotel Manager
Trigger	User clicks “Configure Parameters” after logging in.
Precondition	User must be logged in. Dataset may or may not be uploaded (optional).
Main Flow	1. User sets global min/max multiplier bounds (e.g., 0.7–1.5). 2. User can override bounds per segment (e.g., Travel Agents: 0.85–1.10). 3. User checks “Lock Segment” boxes (e.g., corporate rates fixed). 4. Settings are saved in session and applied during GA execution.
Exception Flow	1a. Max < Min → Display “Maximum bound must be greater than minimum.” 2a. Gov multiplier > 0.8 → Display “Government rate cannot exceed 0.8.”

Table 3.4 Use Case Description for Execute GA Optimization

Use Case ID	UC 004
Use Case	Execute GA Optimization
Purpose	To run the Genetic Algorithm to generate an optimized annual pricing strategy.
Actor	Hotel Manager
Trigger	User clicks “Run Optimization” after configuring parameters.
Precondition	User must be logged in. Dataset must be uploaded.

Main Flow	1. System retrieves dataset and configuration settings. 2. GA engine initializes population (50 chromosomes). 3. Runs 200 generations with tournament selection, arithmetic crossover, Gaussian mutation. 4. Fitness function enforces constraints (Gov cap, multiplier bounds). 5. Best solution is stored in session and displayed in results page.
Exception Flow	1a. No dataset uploaded → Display “Please upload dataset first.” 2a. GA runtime exceeds 5 minutes → Display “Optimization timed out. Try reducing complexity.”

Table 3.5 Use Case Description for View Optimized Pricing Results

Use Case ID	UC 005
Use Case	View Optimized Pricing Results
Purpose	To display the GA-generated pricing strategy and performance metrics.
Actor	Hotel Manager
Trigger	User clicks “View Results” after GA execution.
Precondition	User must be logged in. GA must have completed successfully.
Main Flow	1. System displays monthly pricing table (by room type × segment). 2. Shows comparison: 2024 Actual Revenue vs. GA Optimized Revenue. 3. Displays % uplift and RevPAR metrics. 4. Highlights any constraint violations (if any).
Exception Flow	1a. No GA result found → Display “No optimization results available. Run GA first.”

Table 3.6 Use Case Description for Export Results

Use Case ID	UC 006
Use Case	Export Results

Purpose	To allow the manager to download optimized pricing data for record-keeping or reporting.
Actor	Hotel Manager
Trigger	User clicks “Export CSV” in results page.
Precondition	User must be logged in. GA results must be available.
Main Flow	1. System generates CSV file with columns: Month, Room Type, Segment, Multiplier, Projected Revenue. 2. File is downloaded to user’s device. 3. Confirmation message: “Results exported successfully.”
Exception Flow	1a. No results → Display “Cannot export. Run optimization first.” 2a. Browser blocks download → Display “Enable pop-ups or manually save file.”

3.4.3 Algorithm Design

The pseudocode of the GA-based Hotel Revenue Optimization System can be seen below:

1. Input: Historical data D , target revenue R_target , multiplier bounds $[m_min, m_max]$
2. Classify each month in D into season $S \in \{\text{Lower, Normal, Peak, Super Peak}\}$
3. Initialize population P with 50 random chromosomes
4. Each chromosome $c = [m_1, m_2, \dots, m_48]$ represents multipliers for all (season $\times$ room $\times$ segment) combinations
5. FOR each chromosome c in P DO
6. Compute projected revenue $R_proj(c)$ using D and c
7. IF any Government segment multiplier > 0.8 THEN
8. Set $fitness(c) = 0$ // Invalid solution
9. ELSE
10. $fitness(c) = 1 / (1 + |R_proj(c) - R_target|)$ // Closer to target = higher fitness
11. END IF
12. END FOR
13. WHILE generation < 200 DO
14. Select parents from P using tournament selection
15. Generate offspring via arithmetic crossover and Gaussian mutation

16. Evaluate offspring fitness (as in lines 5–11)
17. Replace weakest individuals in P with offspring
18. END WHILE
19. Output best chromosome c^* and its projected revenue $R_{proj}(c^*)$

The pseudocode above consists of multiple parts:

1. Load real hotel data. The system will first read the dataset that has been entered by the user, in this case it is the Royale Chulan's 2024 dataset.
2. Classify Seasons Automatically. The system will automatically detect which of the months will be classified as Lower (L), Normal (N), Peak (P), and Super Peak (SP).
3. Set target revenue. The responsible hotel manager will enter a desired annual revenue goal
4. Create random pricing strategies. The GA will start with 50 random "pricing strategies". What each of these strategies are is a list of random multipliers for every combination from:

Season * Room type * Customer Segment

5. Test the strategy compared to the target. The system will calculate
 - a. Projected annual revenue
 - b. Rule-breaking
 - c. Proximity towards the projected revenue.
6. Evolve better strategies. The GA will then breed new generations of strategies by selecting ones with better fitness, mixing them and add random small mutations into the next generation.
7. Repeat and improve up to 200 generations.
8. Output the best pricing plan.

3.4.4 User Interface Design

Figure 3.4.4.1 shows the login page for the system

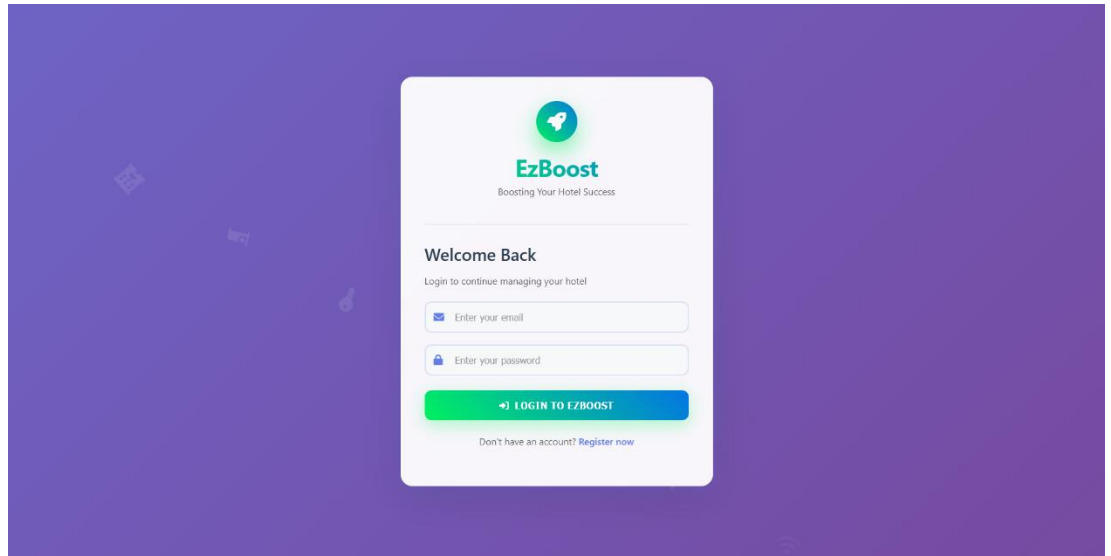


Figure 3.4.4.1 Login.jsp

The login page is a clean and modern design that is straightforward and easy for the users to enter their credentials. From this page, the user can login and navigate into the registration page to register for an account given that they do not have one.

Figure 3.4.4.2 shows the registration page.

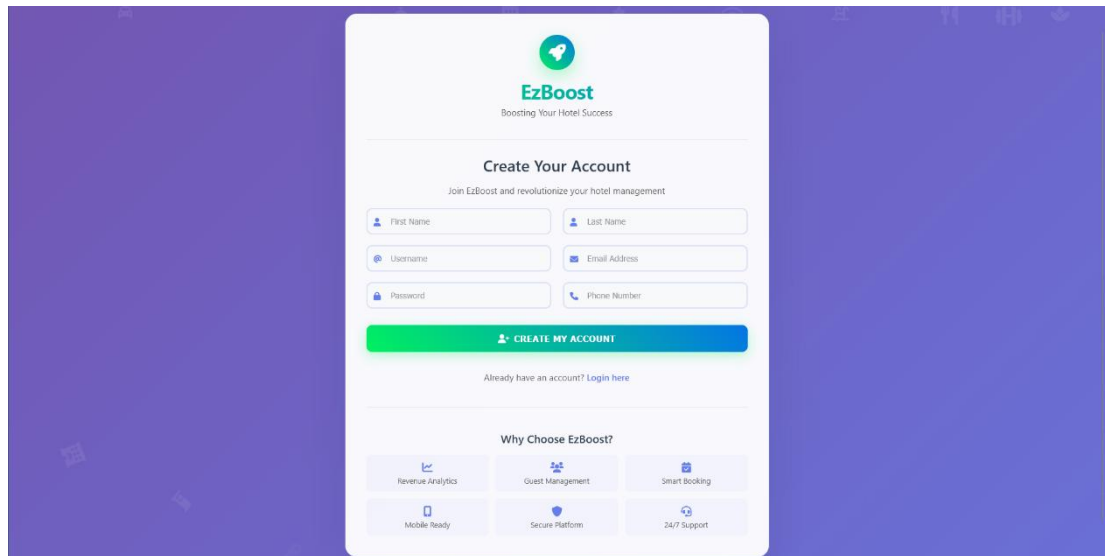
The image shows a registration page for EzBoost. At the top, there is a green circular logo with a white 'E' and the text 'EzBoost' and 'Boosting Your Hotel Success'. Below this is a section titled 'Create Your Account' with the subtitle 'Join EzBoost and revolutionize your hotel management'. There are six input fields: 'First Name', 'Last Name', 'Username', 'Email Address', 'Password', and 'Phone Number'. A green button labeled 'CREATE MY ACCOUNT' is below the fields. A link 'Already have an account? Login here' is below the button. At the bottom, there is a section titled 'Why Choose EzBoost?' with six icons and labels: 'Revenue Analytics', 'Guest Management', 'Smart Booking', 'Mobile Ready', 'Secure Platform', and '24/7 Support'.

Figure 3.4.4.2 Register.jsp

The registration page allows new users to register for the system where it includes input fields for First Name, Last Name, Username, Email, Password, and Phone Number.

Figure 3.4.4.3 shows the dashboard page

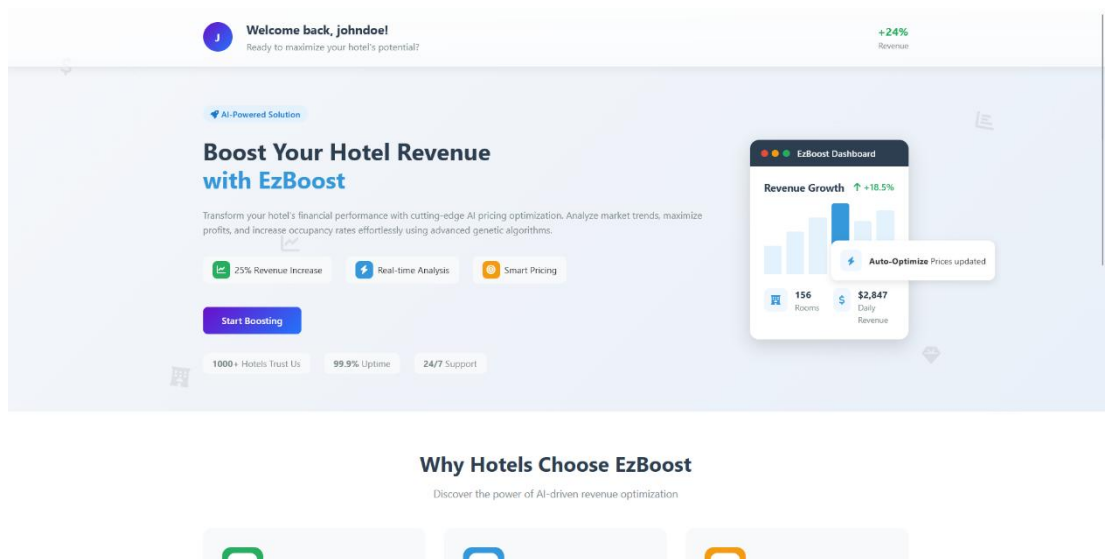


Figure 3.4.4.3 Dashboard.jsp

This is the main dashboard after the user has successfully logged into the website. The website first greets the user by their username, and it highlights the system's core value. It shows that the functionality of the web application is to optimize the revenue of a hotel. In this page, users can navigate through the rest of the web application.

Figure 3.4.4.4 shows the profile page

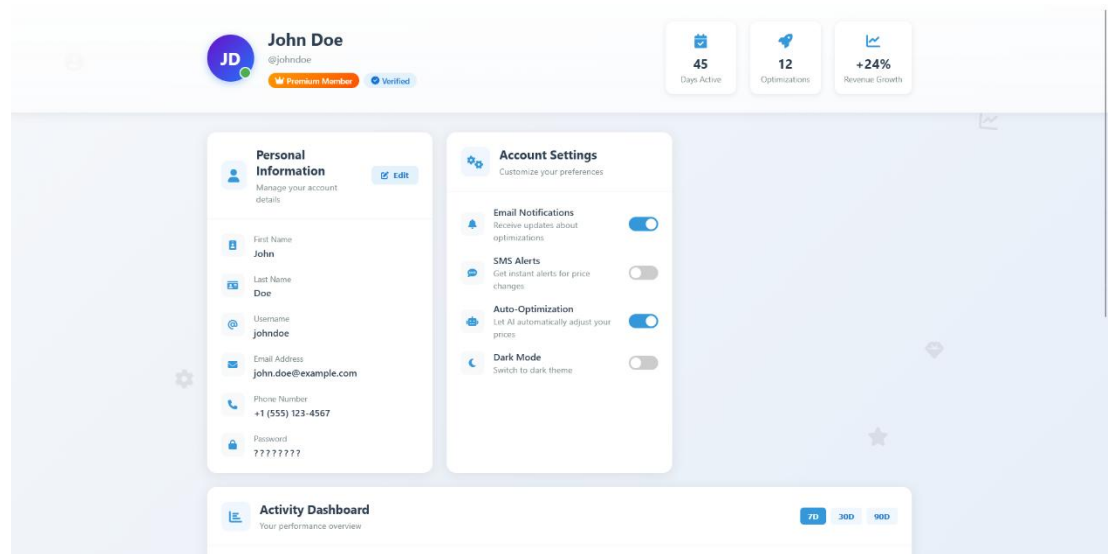


Figure 3.4.4.4 Profile.jsp

This is where users can see and edit their profiles. It includes the user's personal account details and their preferences. This is where the users can customize or edit their profiles.

3.5 Phase 3: Implementation

The development of the web application followed the Model-View-Controller (MVC) architecture that is implemented within a Jakarta EE 10 web application that is built using the Apache NetBeans 27 which is then deployed on the Payara Server 6. All the components in the system were developed to run on standard hardware without any external tools.

3.5.1 Hardware Requirements

The system that has been designed does not require any specialized hardware or any IoT devices. The minimum and the recommended specification are as follows:

Component	Minimum Requirement	Recommended Specification
Processor	Intel Core i3 (2.4 GHz)	Intel Core i5 or higher
RAM	8 GB	16 GB
Storage	10 GB free space	SSD with 20+ GB free
Operating System	Windows 10 / macOS 11 / Linux	Windows 11 / macOS 13 / Ubuntu 22.04 LTS
Network	Localhost (no internet required for core functions)	Stable LAN for multi-user testing (optional)

3.5.2 Software Requirements

The following software tools are required in order to successfully run and operate the system:

Software	Description
Java Development Kit (JDK) 24	Primary programming language for backend logic, GA engine, and data processing. All model classes (GeneticAlgorithmEngine.java, SeasonalityClassifier.java) are written in Java.
Apache NetBeans 27	Integrated Development Environment (IDE) used for coding, debugging, and project management. Supports Jakarta EE 10 and Payara Server integration out of the box.
Jakarta EE 10	Enterprise Java platform used to build the web application. Provides

	standards for Servlets, JSP, and MVC architecture.
Payara Server 6.2025.10	Lightweight, open-source Jakarta EE-compliant application server used to deploy and run the web application locally.
Apache Derby	Embedded relational database used to store Royale Chulan 2024 dataset. Accessed via JDBC without external DBMS.
Jakarta Server Pages (JSP)	Server-side technology used to generate dynamic HTML content for the user interface (e.g., dashboard.jsp, results.jsp).
Bootstrap 5	Frontend CSS framework used to ensure responsive, mobile-friendly UI design across all screens (Login, Dashboard, Profile, etc.).
JavaScript (Vanilla)	Minimal client-side scripting used for form validation (e.g., checking multiplier bounds, required fields). No frameworks (React, Angular) or D3.js are used.
Apache Maven	Build automation tool (managed via NetBeans) for dependency management and WAR file generation.

3.6 Phase 4: Verification and Testing

This phase serves to validate the proposed system meets all the functional requirements and performs within what is acceptable by the users. The testing will focus on correctness, usability, and computational efficiency. The method will be separated into two which are black-box testing for user-facing modules and white-box testing for algorithmic integrity.

3.6.1 Test Case Generation

The functionality testing was conducted using two methods which are black-box for UI interactions and white-box for GA engine validation. The table 3.6.1 presents the test cases conducted.

Case ID	TEST SCENARIO	TEST CASE	PRE-CONDITION	TEST STEPS	TEST DATA	EXPECTED RESULT	POST CONDITION	STATUS (PASS/FAIL)
TC_001	Verify Login Functionality	Verify successful login with valid credentials	User is not logged in. System is running on Payara Server.	1. Enter Username: manager 2. Enter Password: password123 3. Click "Login" button	Username: manager Password: password123	Redirected to Dashboard. Session established.	User is now authenticated. Access to all features granted.	PASS
TC_002	Verify Dataset Upload	Upload Royale Chulan 2024 CSV file	User is logged in. No dataset currently loaded.	1. Click "Upload Dataset" 2. Select royale_chulan_2024.csv 3. Click "Confirm Upload"	File: royale_chulan_2024.csv (valid structure)	System displays: "Dataset uploaded successfully. 12 months loaded." Data stored in Derby tables.	Dataset available for GA execution.	PASS
TC_003	Verify Constraint	Set Government	User is logged in. Dataset	1. Go to "Configure Parameters"	Multiplier bounds: Gov = [0.9, 1.2]	GA output shows	System enforces hard	PASS

	Enforce ment	multipli er > 0.8	uploaded. GA not yet run.	2. Set Gov segment multiplier bound to [0.9, 1.2] 3. Run GA		Gov multiplie rs ≤ 0.8 No solution exceeds cap.	constraint	
TC_ 004	Verify Revenue Uplift	Run GA with default settings	User is logged in. Dataset uploaded.	1. Set global bounds to [0.7, 1.5] 2. Click “Run Optimization” 3. Wait for completion	Default settings, no locked segments	Projected revenue ≥ MYR 21.3M (2024 actual) % uplift displayed (e.g., +8.2%)	Optimize d pricing strategy generated	PASS

Table 3.6.1 Test Cases

3.6.1 Selection of Testers

The testers are selected based on these criteria to ensure that it was evaluated comprehensively across all aspects.

Tester Type	Selection Criteria	Role in Testing
Primary Developer (Self)	Familiar with codebase, GA logic, and system architecture	Takes the role of conducting the white-box testing. The tester validates the internal logic of the system, application of constraint enforcement, and algorithm behaviour. The focus here is to test the functional correctness and the performance of the system.

Peer Evaluator (Computer Science Student)	Has basic web application knowledge and no prior experience on this project.	Conducts black-box testing — uses system as an end-user without knowing backend code. Focus: UI clarity, error handling, and workflow usability. Takes the role of conducting test through the black-box method. The tester here is using the system without knowing the code that is operated. The focus on this tester is to give insights about the UI clarity, error handling, and the usability of the workflow.
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Table 3.6.2 Testers

3.7 Summary

This chapter fully covers the proposed system in terms of the research framework, which outlines the phases of the Web Development Life Cycle (WDLC). The methodology begins by utilizing WDLC as the guide mainly due to its structured, user-centered nature. This makes it well suited to develop this system as it requires transparency and user-friendliness to serve as a hotel revenue optimization tool. The project framework is established to respectively map each of the research objectives that ranges from identifying GA-based pricing features until evaluating the system's performance to specific WDLC phases. WDLC starts off with the requirements phase which consists of details on how information was gathered via literature reviews. The design of the

system utilizes the Model-View-Controller (MVC) architecture within the Apache NetBeans IDE. Verification of the system was done through both black-box and white-box testing to validate its functionality. Together, the phases of the WDLC ensured that the final system is technically robust while staying practically relevant for the hoteliers to integrate.

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